

Make sense of dealer-talk

ACPI: Advanced Configuration and Power Interface is a power management specification that allows the OS to control the power distributed to the computer's devices.

AGP: Accelerated Graphics Port is a PCI-based interface that was designed for demands of 3D graphics applications. The 32-bit AGP channel directly links the graphics controller to the main memory. While the channel runs at only 66 MHz, it supports data transmission during both the rising and falling ends of the clock cycle, yielding an effective speed of 133 MHz.

BIOS: Basic Input and Output System program resides in the ROM chip, and provides basic instructions for controlling the hardware. Both, the OS and application software, use BIOS routines to ensure compatibility.

Bus: This is a data pathway. The term is used especially to refer to the connection between the processor and system memory, and between the processor and PCI or ISA local buses. Local buses are those that operate within a single system (as opposed to a network bus, which connects multiple systems).

Cache: This is a temporary, fast storage area that holds data from a slower storage device for quick access as needed by an application. Access time is fast using a cache, because the needed information is stored in the SRAM instead of in the slower DRAM. Note that the cache is also much smaller than your regular memory: a typical cache size is 512 KB, while you may have as much as 2 GB of regular memory.

Chipset or core logic: Two or more integrated circuits which

control the interfaces between the system processor, RAM, I/O devices and adapter cards.

CMOS: Complementary Metal-Oxide Semiconductors are chips that hold the basic start-up information for the BIOS.

Form factor: This is an industry term for the size, shape, power supply type and external connector type of the PCB (personal computer board) or motherboard. The standard form factors are the AT and ATX, although TYAN also makes some Baby-AT boards.

FSB (Front Side Bus): The FSB connects CPU to main memory (RAM) and is a critical component in determining the speed of your computer.

IDE: Integrated Device Electronics is a simple, self-contained hard drive interface, that is most commonly used in desktop systems.

ISA: The Industry Standard Architecture is a slower 8-bit or 16-bit Bus that is only used with very old hardware.

PCI: The Peripheral Component Interconnect is a 32-bit local bus, faster than ISA, and the current standard for add-ons such as network and sound cards.

RDRAM: Developed by Rambus, Inc, RDRAM is the fastest memory technology used. While the more common SDRAM delivers data at a maximum speed of 133 MHz, RDRAM currently transfers data at up to 1200 MHz.

UltraDMA/33/66/100: This is a fast version of the old DMA channel. UltraDMA is also called UltraATX. Without UltraDMA controller, your system cannot take advantage of the higher data transmission rates of the new UltraDMA/UltraATA HDDs.

Tips On Installing Motherboards

The motherboard is the physical and electronic foundation of your system. Here are a few tips that will help smoothen the installation process:

Get it set right

Before you install the motherboard, consult the accompanying manual. You may have installed motherboards numerous times, but manufacturers have addenda that describe special considerations or last-minute changes incorporated into the board. Also, mark all the jumper settings in the manual tables so that they correspond to your processor, RAM and other system components.

Handle with caution

The motherboard has very sensitive electronic components that are susceptible to damage by static charge. Therefore, ensure that you are properly grounded and that the motherboard is kept in its anti-static packaging until the time of installation.

Set the jumpers

Set the jumpers to their correct positions before installing the motherboard into the system cabinet. It's far easier to access certain jumpers when the board is outside the system.

Mount the processor, heatsink and RAM beforehand. It can be quite difficult to mount the processor after the board is seated in the cabinet; since the power and IDE cables get in the way. This also allows you to visually verify if the heatsink is properly seated on the processor. Remember to connect the CPU fan to the correct point on the motherboard.

Ensure a comfortable fit

The power conditioning capacitors next to the processor socket can

get in the way of mounting the CPU heatsink.

Make sure your heatsink does not push against these, as this could damage or dislodge them from the motherboard. Also, if you have a special heatsink, make sure it does not push against other surface-mounted components such as capacitors next to the processor socket.

Anchor the board

Make sure that your motherboard is comfortably seated in the cabinet. Ideally your cabinet should have a detachable backplate, which lets you mount the motherboard using the spacers and mounting screws outside the system. Ensure that the motherboard has anchor points near all the extremities of the board so that it's firmly seated and all edges are adequately supported.

Tighten later

Most of the time, one would tighten all the screws of the motherboard on placing it into the system. A better option is to lightly fasten all the mounting screws so that there's still a bit of play in the motherboard after it has been mounted. Tighten the motherboard mounting screws after the rest of the add-on cards have been placed in their respective PCI or AGP slots. This eliminates any unnecessary use of force to get cards into a slot.

Attach the IDE cables

The IDE cables are usually situated near one edge of the motherboard. Make sure the board is not bent when plugging in the ends of the cables. If you're not careful, this could cause breakages in the delicate tracks on the motherboard surface, rendering it useless.

